



An official website of the United States government

[Here's how you know](#)

FULL TEXT LINKS



[Carbohydr Polym.](#) 2020 May 1:235:115969. doi: 10.1016/j.carbpol.2020.115969.

Epub 2020 Feb 10.

Structural characterization, antioxidant and immunomodulatory activities of a neutral polysaccharide from Cordyceps militaris cultivated on hull-less barley

Ying Zhang ¹, Yan Zeng ², Yongsheng Cui ³, Hanmin Liu ⁴, Caixia Dong ⁵, Yuanxia Sun ⁶

Affiliations

PMID: 32122503 DOI: [10.1016/j.carbpol.2020.115969](#)

Abstract

A water-soluble neutral polysaccharide (SDQCP-1) was isolated from fruit bodies of Cordyceps militaris cultivated on hull-less barley. SDQCP-1 was composed of mannose, glucose and galactose in the mole ratio of 13.3:1.0:9.7, with an average molecular weight of 19.3 kDa. Based on results from methylation analysis, GC-MS and NMR, SDQCP-1 was elucidated to be a glucogalactomannan with a backbone composed of (1→2)-α-D-Manp (48.4 %) and (1 → 4)-β-D-Glcp (1.2 %) residues. Its side chains were branched at O-6 position of (1→2)-α-D-Manp mainly by (1 → 2)-β-D-Galp or (1 → 6)-α-D-Manp residues which were terminated mainly with α-D-Galp, α-D-Galp residues. Besides exhibiting a good antioxidant capacity with an ORAC_{FL} value of 24.7 mmol Trolox/g and a TEAC value of 202.4 μmol Trolox/g, SDQCP-1 also could stimulate macrophages to release NO, TNF-α, IL-6, and IL-10, and mainly induced M1 polarization of macrophages. The findings indicated that SDQCP-1, from C. militaris cultivated on hull-less barley, could be used as potential natural antioxidant and immunomodulator in functional foods or medicine.

Keywords: Antioxidant activity; Cordyceps militaris; Hull-less barley; Immunomodulatory activity; Polysaccharide.

Copyright © 2020 Elsevier Ltd. All rights reserved.

[PubMed Disclaimer](#)

Related information

[MedGen](#)

[PubChem Compound \(MeSH Keyword\)](#)

LinkOut – more resources

Full Text Sources

[Elsevier Science](#)

Medical

[MedlinePlus Health Information](#)

Miscellaneous

[NCI CPTAC Assay Portal](#)